William Brower

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EDUCATION

Northwestern University | GPA: 3.68 | *Computer Engineering Major / Artificial Intelligence Minor*
Relevant classes: Public Speaking, Digital Circuit Design, Data Structures, Foundations of Optimization, Statistical Systems, Fundamentals of Computer Programming, Physics (Mechanics, Electromagnetism, Optics), 3D/Multivariable Calculus, Computer System Software, Differential Equations, Linear Algebra

Holy Family High School *Class of 2024 Valedictorian*
Presidential Honors, Student Ambassador, UGPA: 3.96, WGPA: 4.56, Class Rank: 1/211

PROJECTS

Personal Website
Independently created a website front-end that contains multiple pages and responds to user input, hosted on GitHub using React, coded in HTML, CSS, and Typescript. Currently integrating a backend using Node.js.

Debugger for Python
Coded a debugger for Python in C that uses tokenization to process Python code and uses dynamically allocated RAM for the values contained in the Python code.

Electrical Pill Box Design
Worked with a team to design a pill box that sends daily text message reminders to the client to take their medicine. Independently wrote the C++ code that controlled these functions through an Arduino (ESP32).

Video Game
Independently developed a pong-style game using ARM Assembly that updates based on user input and directly interacts with the architecture of a DE1SoC-style computer system.

EXPERIENCE

University of Colorado Anschutz Medical Campus *Research Assistant, June - August 2023*

- One of only 12 students in the state of Colorado selected to perform research at CU Anschutz. I worked full-time with three post-docs, researching chronic and polycystic kidney disease. Organized and presented key results during a poster session and documented lab progress in Excel.
- Designed a poster that outlined the basic results of the research, and used this poster in a public presentation.
- Performed statistical calculations on the data, including chi-squared tests, standard deviation, correlation coefficient (r-values), F-tests, Z-scores, T-tests, and p-values.

Northwestern University: Image and Video Processing Lab *Research Assistant, Mar. 2026 - Present*

- Currently work as a student researcher, helping to develop new methods and models for computer vision and artificial intelligence technology in various sources of media (images, videos, social media posts, etc).
- Write new code for this purpose using Python with OpenCV and Jupyter Notebook.

Teacher's Assistant *Oct. 2023 - Jun. 2024*
Assisted US/Comparative Government teacher by creating outlines that every student used to study for tests. I used PowerPoint and Word to design materials for every class, outlining key ideas and concepts.
Created graphs and figures to communicate complex ideas or trends in political data to students.

Step Up Youth *Volunteer, Feb. 2020 - Mar. 2026*
Volunteered over 500 hours assembling and distributing concessions for a non-profit organization, where my earnings benefit underprivileged youth in the local Denver community. In addition, a portion of my earnings directly supports my college tuition. Through this, I help both my community and support my education.

ADDITIONAL INFO

Coding & Technical Skills

- JavaScript/TypeScript, HTML, and CSS; React/Node.js; Verilog with Xcelium and Genus; MATLAB; C++ and C; Python; SQL querying and syntax; Linux, and GitHub integration into project development using Docker containers, and various industry testing standards for written code (Google Test for C, etc.).
- Knowledge of Python data processing libraries/tools (pandas, numpy, scikit-learn, matplotlib, opencv) and data fitting/categorization (k-means clustering, least-squares regression, random/decision trees, Naive Bayes, and dimensionality reduction).

Misc.

- Fluent in English, Proficient in Spanish, US Citizen, US Work Authorized
- 2024 Coca-Cola Scholar Semi-Finalist, National Hispanic Scholar, AP Scholar with Distinction, 1560 SAT